Polynomial Long Division and Synthetic Division

Course: Pre-Calculus-I

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The Division Algorithm for Polynomials

Theorem (Division Algorithm)

If $P(x)$ and $D(x)$ are polynomials with $D(x) \neq 0$ and $\deg D \leq \deg P$, then there exist unique polynomials $Q(x)$ (quotient) and $R(x)$ (remainder) such that

$$P(x) = D(x)\,Q(x) + R(x),$$

where either $R(x) = 0$ or $\deg R < \deg D$.

- P = dividend, D = divisor, Q = quotient, R = remainder.
- If $R(x) = 0$, then $D(x)$ is a factor of $P(x)$.

Note

This mirrors integer division: $17 = 5 \cdot 3 + 2$, *with remainder* $2 < 5$.

Remainder and Factor Theorems

Remainder Theorem

If $P(x)$ is divided by $(x - c)$, then the remainder is $P(c)$.

Factor Theorem

$(x - c)$ is a factor of $P(x) \iff P(c) = 0$.

Quick example: $P(x) = x^3 - 4x + 6$ divided by $(x - 2)$.

$$P(2) = 8 - 8 + 6 = 6 \Rightarrow \text{remainder } 6.$$

Since $P(2) \neq 0$, $(x - 2)$ is not a factor of $P(x)$.

Long Division: Step-by-Step Method

Algorithm:

1. Write P and D in descending powers of x; fill missing terms with 0.
2. Divide the leading term of P by the leading term of D; this is the next quotient term.
3. Multiply that term by D and subtract from P.
4. Repeat with the new remainder until its degree is less than $\deg D$.

Note

Subtracting a polynomial is the same as adding its negative — be careful with signs.

Worked Example 1: Long Division

Divide $P(x) = 2x^3 - 3x^2 + 4x - 5$ by $D(x) = x - 2$.

$$
\begin{array}{r|l}
 & 2x^2 + x + 6 \\
\hline
x - 2 & 2x^3 - 3x^2 + 4x - 5 \\
 & -(2x^3 - 4x^2) \\
\hline
 & x^2 + 4x - 5 \\
 & -(x^2 - 2x) \\
\hline
 & 6x - 5 \\
 & -(6x - 12) \\
\hline
 & 7
\end{array}
$$

Result:

$$\frac{2x^3 - 3x^2 + 4x - 5}{x - 2} = \underbrace{2x^2 + x + 6}_{Q(x)} + \underbrace{\frac{7}{x - 2}}_{\frac{R(x)}{D(x)}}.$$

Worked Example 2: Long Division with Missing Terms

Divide $P(x) = x^4 - 1$ by $D(x) = x^2 + 1$.

Rewrite with zero coefficients: $P(x) = x^4 + 0x^3 + 0x^2 + 0x - 1$.

$$
\begin{array}{r|l}
 & x^2 - 1 \\
\hline
x^2 + 1 & x^4 + 0x^3 + 0x^2 + 0x - 1 \\
 & -(x^4 + x^2) \\
\hline
 & \quad - x^2 + 0x - 1 \\
 & -(-x^2 - 1) \\
\hline
 & \quad 0
\end{array}
$$

Result:

$$x^4 - 1 = (x^2 + 1)(x^2 - 1) + 0.$$

So $x^2 + 1$ divides $x^4 - 1$ exactly.

Synthetic Division: When and Why

When to Use

Synthetic division is a shortcut for dividing by $x - c$ (a linear divisor with leading coefficient 1).

Advantages:

- Much faster than long division.
- Directly gives $P(c)$ (remainder) and quotient coefficients.

Setup:

1. Write coefficients of P (descending powers, include zeros).
2. Write c to the left (from divisor $x - c$).
3. Bring down the first coefficient; multiply by c; add.

Note

Divide by $x + 3$ means $c = -3$ (since $x + 3 = x - (-3)$).

Synthetic Division: Layout

Divide $P(x) = 2x^3 - 3x^2 + 4x - 5$ by $x - 2$ (so $c = 2$).

2	2	-3	4	-5
		4	2	12
	2	1	6	**7**

Reading the result:

- Quotient: $2x^2 + x + 6$ (one degree less than P).
- Remainder: 7.

$$\frac{2x^3 - 3x^2 + 4x - 5}{x - 2} = 2x^2 + x + 6 + \frac{7}{x - 2}.$$

Matches the long division from the previous section. ✓

Synthetic Division with $c < 0$

Divide $P(x) = x^3 + 2x^2 - 5x + 1$ by $x + 2$ (so $c = -2$).

$$\begin{array}{r|cccc} -2 & 1 & 2 & -5 & 1 \\ & & -2 & 0 & 10 \\ \hline & 1 & 0 & -5 & \mathbf{11} \end{array}$$

Result:

$$x^3 + 2x^2 - 5x + 1 = (x + 2)(x^2 - 5) + 11.$$

Check with Remainder Theorem:

$$P(-2) = -8 + 8 + 10 + 1 = 11. \checkmark$$

Synthetic Division to Test Factors

Question: Is $x-3$ a factor of $P(x)=x^3-4x^2+x+6$?

$$\begin{array}{r|cccc} 3 & 1 & -4 & 1 & 6 \\ & & 3 & -3 & -6 \\ \hline & 1 & -1 & -2 & \mathbf{0} \end{array}$$

Remainder $=0$, so $(x-3)$ is a factor:

$$P(x)=(x-3)(x^2-x-2)=(x-3)(x-2)(x+1).$$

Roots: $x=3,\ 2,\ -1$.
All real, so the graph crosses the x-axis at all three points.

Summary: Long vs. Synthetic Division

Feature	Long Division	Synthetic Division
Divisor	any polynomial	$x - c$ only
Speed	slow	fast
Works with	all P, D	coefficients only
Gives remainder	yes	yes
Shows full algebra	yes	compact

Note

Use synthetic division whenever the divisor is linear of the form $x - c$; use long division otherwise.

Class Practice Questions

P1. Use long division to divide $3x^3 + x^2 - 4x + 5$ by $x + 1$.
P2. Divide $x^4 - 3x^2 + 2$ by $x^2 - 1$ using long division.
P3. Use synthetic division to divide $2x^3 - 7x^2 + 4x - 1$ by $x - 3$.
P4. Is $x + 2$ a factor of $x^3 + 4x^2 + x - 6$? Justify with synthetic division.
P5. Use synthetic division to evaluate $P(4)$ for $P(x) = x^3 - 2x^2 - 5x + 6$.